Fundamental Theorem of Arithmetic

An End-to-End Meta-Math Walkthrough

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Abstract

Existence uses strong induction. Uniqueness uses a different branch: Bézout gives Euclid's lemma, which matches and cancels prime factors one at a time. This report presents a complete proof relative to explicit imports, a dependency graph, versioned node contracts, five-component activities, exact examples, and an illustrative repair. The interactions are authored explanations, not a transcript of deployed agents or a proof-assistant certificate.

Project NT-02 Report version 1 Number Theory

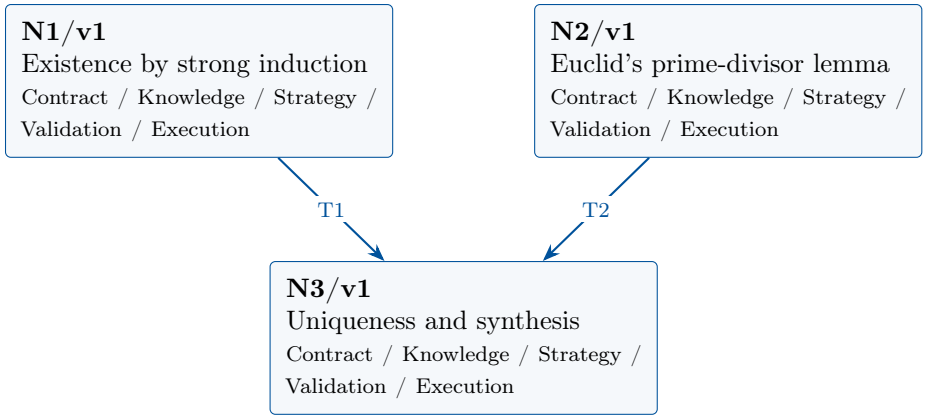
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Every integer $n \geq 2$ can be written as a finite product of primes. If $n = p_1 \cdots p_r = q_1 \cdots q_s$ with all factors prime, then $r = s$ and the two lists agree after a permutation, including multiplicities.*

A. Foundations and permitted imports

1. A prime is an integer $p \geq 2$ whose only positive divisors are $1, p$. A composite integer $n \geq 2$ has a divisor a with $1 < a < n$, and hence $n = ab$ with $1 < a, b < n$. Strong induction on positive integers is foundational.
2. NT-01/N5, v1: for integers x, y not both zero, there exist integers u, v with $ux + vy = \gcd(x, y)$. In particular, if a prime p does not divide x, then $\gcd(p, x) = 1$. Unique factorization is not an import.

II. NODE INTERACTION WORKFLOW



Blue arrows are accepted mathematical dependencies, with exact transfers T1–T2 below. Repair messages are described separately; they are not premises of the proof.

A. Versioned evidence-transfer ledger

T1: N1/E1 v1 to N3.3
Claim. Every integer $n \geq 2$ has a finite factorization into primes.
Compatibility. N1 guarantees a prime product exists; uniqueness compares any two such products.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T2: N2/E2 v1 to N3.1
Claim. For a prime p, $p \mid xy$ implies $p \mid x$ or $p \mid y$; a prime dividing a finite nonempty product divides a factor.
Compatibility. All compared factors are prime, so finite-product divisibility applies.
Evidence. Node N2 activities below and the continuous proof in Section IV.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Existence by strong induction

Object and role. Build a finite prime product without using uniqueness.

Inputs. $n \geq 2$; prime/composite definitions; strong induction.

C — Contract

Export existence only, preserving a separate uniqueness obligation.

K — Knowledge

Retain the induction hypothesis for integers $2 \leq t < n$.

S — Strategy

Split a composite integer into two strictly smaller factors.

V — Validation

The factor 1 is not prime and cannot provide a decreasing composite split.

E — Execution

The base $n = 2$ is prime. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N1.1 The base $n = 2$ is prime.

N1.2 If n is prime, use the one-factor product.

N1.3 Otherwise write $n = ab$ with $1 < a, b < n$; concatenate the two products supplied by induction.

Export E1/v1. Every integer $n \geq 2$ has a finite factorization into primes.

Acceptance gate. Every integer $n \geq 2$ falls into a prime case or a composite case with strictly smaller inputs.

Targeted repair. If a factor equals 1 or n, select a proper divisor before invoking induction; recheck N3.

B. N2/v1: Euclid's prime-divisor lemma

Object and role. Provide the uniqueness bridge using Bézout, without circular use of factorization.

Inputs. p prime, $p \mid xy$; NT-01/N5 Bézout.

C — Contract

Restrict the export to prime divisors and record the Bézout import.

K — Knowledge

Retain $p \nmid x \Rightarrow \gcd(p, x) = 1$ and an integer Bézout certificate.

S — Strategy

If the first factor is not divisible by p, transfer divisibility to the second using a linear combination.

V — Validation

The statement fails for composite 6 dividing $2 \cdot 3$.

E — Execution

If $p \mid x$, the conclusion already holds. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N2.1 If $p \mid x$, the conclusion already holds.

N2.2 Otherwise write $up + vx = 1$ by Bézout and multiply by y.

N2.3 Since $p \mid upy$ and $p \mid vxy$, conclude $p \mid y$; induct on the number of factors for a finite product.

Export E2/v1. For a prime p, $p \mid xy$ implies $p \mid x$ or $p \mid y$; a prime dividing a finite nonempty product divides a factor.

Acceptance gate. Both cases close and the finite-product extension uses induction on its length.

Targeted repair. Restore the prime hypothesis in an overgeneralized candidate and recheck N3 and every external consumer.

C. N3/v1: Uniqueness and synthesis

Object and role. Match prime factors, cancel in the integers, and join uniqueness to existence.

Inputs. N1/E1, N2/E2; two finite prime products for the same $n \geq 2$.

C **Contract**

Publish existence and uniqueness together only after multiplicities and exhaustion are handled.

K **Knowledge**

Retain the two prime lists, Euclid's lemma, and integer cancellation by a nonzero factor.

S **Strategy**

Remove one matched prime from each list and decrease the remaining length.

V **Validation**

A leftover nonempty prime product cannot equal 1; repeated primes must be matched repeatedly.

E **Execution**

Euclid's lemma gives $p_1 \mid q_j$ for some j, so primality forces $p_1 = q_j$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N3.1 Euclid's lemma gives $p_1 \mid q_j$ for some j, so primality forces $p_1 = q_j$.

N3.2 Permute the second list and cancel the common nonzero integer.

N3.3 Repeat with shorter lists; both must exhaust together, since every prime is at least 2.

Export E3/v1. Prime factorization exists and its multiset of factors is unique.

Acceptance gate. The matching terminates with equal lengths and multiplicities; existence comes from N1.

Targeted repair. If multiplicities are lost, restore lists rather than sets and recheck the cancellation induction.

IV. CONTINUOUS MATHEMATICAL PROOF

Proof. We first prove existence by strong induction (N1). The integer 2 is prime. For $n > 2$, a prime n is already a one-factor product. Otherwise $n = ab$ for integers $1 < a, b < n$. Applying the induction hypothesis separately to a and b and concatenating the finite products gives a prime factorization of n.

For uniqueness, prove Euclid's lemma independently (N2). Let p be prime and $p \mid xy$. If $p \mid x$, there is nothing to prove. Otherwise the positive common divisors of p, x consist only of 1, so Bézout (NT-01) gives $up + vx = 1$. Multiplication by y yields

$$y = upy + vxy.$$

Both summands are divisible by p, hence $p \mid y$. Induction on the number of factors extends this to any nonempty finite product.

Now suppose $p_1 \cdots p_r = q_1 \cdots q_s$ (N3). The prime p_1 divides some q_j by the lemma. Since q_j is prime and $p_1 > 1$, they are equal. Move q_j to the front and cancel this nonzero common factor in the integer equality. Repeating decreases the length of the first list. If one list were exhausted earlier than the other, an empty product 1 would equal a nonempty product of integers at least 2, a contradiction. Consequently the lists have equal length and matching factors, with repetitions counted. This proves uniqueness up to permutation. Together with the independent existence argument it proves the theorem.

□

V. WORKED ORCHESTRATION

Existence can split $360 = 36 \cdot 10 = (2 \cdot 3)^2(2 \cdot 5)$, giving

$$360 = 2^3 3^2 5.$$

For any competing prime product, N2 forces a factor 2. N3 cancels it and compares products for 180; repeating gives 90, then 45. Two forced cancellations of 3 leave 5, and cancelling 5 leaves 1. An additional factor would make the remaining product at least 2. This trace exhibits the multiplicities $3, 2, 1$; the general proof establishes the claim for all integers.

A. Boundary and hypothesis audit

- $n = 2$ closes with one factor. The integer 1 may be represented by an empty product, but is outside the theorem's stated domain.
- Primes cannot be replaced by arbitrary positive divisors: $6 \mid 2 \cdot 3$ while $6 \nmid 2$ and $6 \nmid 3$.
- Integer cancellation here requires only a nonzero factor; it is not the modular cancellation rule requiring a unit.

VI. ILLUSTRATIVE FAILURE AND REPAIR

This is a deliberately authored teaching trace. A diagnostic asks for correction; it cannot prove a predecessor by importing its descendant's conclusion. A revised export must be checked before its consumers are reused.

Candidate. N2 proposes $d \mid xy \Rightarrow d \mid x$ or $d \mid y$ for arbitrary $d > 1$.

Detection. Validation supplies $d = 6, x = 2, y = 3$. Knowledge observes that Bézout with $x = 2$ gives gcd 2, not 1.

Correction. Strategy restores the prime-divisor contract. Execution derives $y = upy + vxy$ under $p \nmid x$.

Recheck. Withhold the overgeneralized candidate; recheck N3 and the CRT, Fermat, and square-factor uses that consume the lemma.

Publication. Contract publishes the prime lemma, not the false composite extension. The original factorization theorem is unchanged.

VII. REUSABLE RESULT AND PROVENANCE

Reusable exports retain each node's version 1 statement, assumptions, and proof. The final export is the theorem in Section 1, proved relative to its declared imports.

Source record. Adapted from the original project Markdown:
`fundamental-theorem-of-arithmetic.md`

Typography, workflow conventions, and component cards follow the workspace Meta-Math manuscript. This authored exposition does not claim formal certification.